

Experiment No. 14

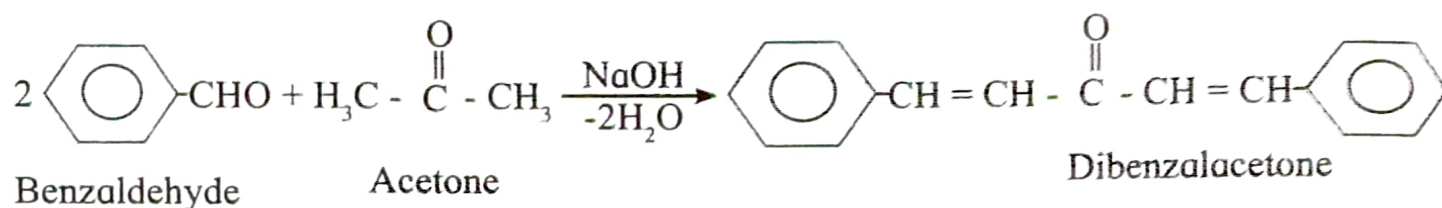
Organic Preparations

Date: / /

Aim: To prepare dibenzalacetone.

Theory : Dibenzalacetone is prepared by the condensation of benzaldehyde with acetone in the presence of sodium hydroxide. It is an example of *Claisen-Schmidt reaction*.

Chemical Reaction: The chemical equation can be written as:



Apparatus: Conical flask, beaker, test tube, funnel, filter paper, etc.

Chemicals: Benzaldehyde, acetone, 10% NaOH, ethanol/rectified spirit, ice.

Procedure:

1. Take 0.5 mL benzaldehyde, 0.2 mL acetone and 5 mL methylated spirit in a small conical flask. Cork the conical flask and shake well.
2. Add 1 mL of 10% NaOH drop wise to the conical flask with constant shaking.
3. Cork the conical flask and shake vigorously for about 10 minutes.
4. Intermittently remove the cork of the flask to release the pressure developed.
5. Keep it for 20 minutes to complete the reaction and then cool it in ice cold water.
6. Filter the yellow coloured solid and wash it with water.
7. During recrystallization dissolve the solid in hot rectified spirit, cool, filter and dry the crystals. (Do not heat directly with burner, use hot water bath)

Result: 1. Yield of dibenzalacetone = 3.3 g
2. Colour of dibenzalacetone crystal = Yellow

Remark and sign of teacher:

MCQ

Select the correct answer for the following multiple choice type questions.

1. The colour of dibenzalacetone is.....
 a. white b. green ☒ c. yellow d. red
2. The process used in the formation of dibenzalacetone is.....
~~a. condensation~~ b. addition
 c. oxidation d. double decomposition
3. Dibenzalacetone contains
 a. two phenyl groups ☒ b. two benzyl rings
 c. one benzene ring d. two ketonic groups
4. Claisen-Schmidt reaction takes place between
☒ a. aliphatic ketone and aromatic aldehyde
 b. aliphatic ketone and aromatic ketone
 c. aromatic ketone and aromatic aldehyde
 d. aromatic ketone and aliphatic aldehyde
5. Recrystallisation of dibenzalacetone is carried out by using
 a. NaOH ☒ b. ethanol c. water d. acetone

Short answer questions

1. Write the molecular formula of dibenzalacetone.

Ans. $C_{17}H_{14}O$

2. Which reagent is used for condensation of benzaldehyde and acetone?

Ans. benzaldehyde and acetone condensed in presence of sodium hydroxide

3. Write the name reaction for preparation of dibenzalacetone.

Ans. Claisen - Schmidt Reaction

4. Is it necessary to recrystallize the dibenzalacetone obtained from the reaction? Explain Why?

Ans. Yes, because that we get pure crystals of the compound.

5. What will be the product obtained when one equivalent of benzaldehyde reacts with one equivalent of acetone?

Ans. when two equivalent of benzaldehyde and one equivalent of acetone reacts, one equivalent of Dibenzalacetone is prepared

Remark and sign of teacher: